

z-transform

z-transform is the conversion of time domain / discrete signals into z domain for analysis and manipulation, given by the formula:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}, \quad z \text{ is a complex variable.}$$

82B4

z : PF method : $\frac{1}{3} < |z| < 2$
 $3z^2 - 4z + 1$

Here,

$$H(z) = \frac{z}{3z^2 - 4z + 1}$$

$$= \frac{A}{z - \frac{1}{3}} + \frac{B}{z - 1}$$

$$A = \frac{z}{z - 1} \bigg|_{z = \frac{1}{3}} = \frac{3}{2}$$

$$B = \frac{z}{3z^2 - 4z + 1} \bigg|_{z = 1} = \frac{-1}{2}$$

$$\therefore, H(z) = \frac{1}{2} \frac{z}{z - 1}$$

$$H(z) = \frac{z}{3z^2 - 4z + 1}$$

$$= \frac{z/3}{z^2 - 4/3 z + 1/3}$$

$$= \frac{A}{z - 1} + \frac{B}{z - 1/3}$$

$$A = \frac{z/3}{z - 1/3} \bigg|_{z = 1} = \frac{1}{2}$$

$$B = \frac{z/3}{z - 1} \bigg|_{z = 1/3} = \frac{-1}{6}$$

$$H(z) = \frac{z}{3z^2 - 4z + 1}$$

$$= \frac{z/3}{z^2 - 4/3 z + 1/3}$$

$$\text{or } H = \frac{1/3}{z^2 - 4/3 z + 1/3}$$

$$= \frac{A}{z-1} + \frac{B}{z-1/3}$$

$$A = \frac{1/3}{z-1/3} \Big|_{z=1}$$

$$= \frac{1}{2}$$

$$B = \frac{1/3}{z-1} \Big|_{z=1/3}$$

$$= -\frac{1}{2}$$

so,

$$H(z) = \frac{1}{2} \frac{1}{z-1} - \frac{1}{2} \frac{1}{z-1/3}$$

$$H(z) = \frac{1}{2} \frac{z}{z-1} - \frac{1}{2} \frac{z}{z-1/3}$$

for $1/3 < |z| < 1$:

$$h(n) = \text{IDTFT} = -\frac{1}{2} (1)^n u(-n-1) + -\frac{1}{2} \left(\frac{1}{3}\right)^n u(n)$$

if $|z| > 1$:

$$h(n) = \frac{1}{2} (1)^n u(n) - \frac{1}{2} \left(\frac{1}{3}\right)^n u(n)$$

if $|z| < 1/3$:

$$h(n) = -\frac{1}{2} (1)^n u(n-1) + \frac{1}{2} \left(\frac{1}{3}\right)^n u(-n-1)$$

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$$X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$$

$$= \frac{z^2}{z^2 - 1.5z + 0.5}$$

$$\frac{X(z)}{z} = \frac{z}{(z - 0.5)(z - 1)}$$

$$= \frac{A}{z-1} + \frac{B}{z-0.5}$$

$$A = \frac{z}{z-0.5} \bigg|_{z=1} \quad B = \frac{z}{z-1} \bigg|_{z=0.5}$$

$$= 2 \quad = -2$$

$$X(z) = \frac{2z}{z-1} - \frac{z}{z-0.5}$$

for $\frac{1}{2} < |z| < 1$:

$$x(n) = -2(1)^n u(-n-1) - (0.5)^n u(n)$$

if $|z| > 1$:

$$x(n) = 2(1)^n u(n) - (0.5)^n u(n)$$

if $|z| < 0.5$ $\sim n-1$

$$x(n) = -2(1)^n u(n) + (0.5)^n u(-n-1)$$

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3 The region of z plane, where the z -transform of a discrete-time signal converges is called its region of convergence.

$$X(z) = \frac{2z^3 - 5z^2 + z + 3}{(z-1)(z-2)} \quad |z| < 1$$

$$\begin{aligned} \frac{X(z)}{z} &= \frac{2z^3 - 5z^2 + z + 3}{z^2 - 3z + 2} \\ &= \frac{2z^2 -}{z^2 - 3z + 2} \\ &= \frac{2z(z^2 - 3z + 2) + z^2 - 3z + 3}{z^2 - 3z + 2} \\ &= 2z + \frac{z^2 - 3z + 2 + 1}{z^2 - 3z + 2} \\ &= 2z + 1 + \frac{1}{(z-1)(z-2)} \rightarrow \text{let } X_1 \end{aligned}$$

$$X_1(z) = \frac{1}{(z-1)(z-2)}$$

$$\frac{X_1(z)}{z} = \frac{A}{z} + \frac{B}{z-1} + \frac{C}{z-2}$$

$$\begin{aligned} A &= \frac{1}{(z-1)(z-2)} \Big|_{z=0} & B &= \frac{1}{z(z-2)} \Big|_{z=1} & C &= \frac{1}{z(z-1)} \Big|_{z=2} \\ &= \frac{1}{2} & &= -1 & &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{so, } \frac{X_1(z)}{z} &= \frac{1}{z(z-1)(z-2)} = \frac{1}{2} \cdot \frac{1}{z} - \frac{1}{z-1} + \frac{1}{2} \cdot \frac{1}{z-2} \\ X_1(z) &= \frac{1}{2} - \frac{z}{z-1} + \frac{1}{2} \cdot \frac{z}{z-2} \end{aligned}$$

$$\text{so, } X(z) = 2z + 1 + \frac{1}{2} - \frac{z}{z-1} + \frac{1}{2} \cdot \frac{z}{z-2}$$

$$= 2z + \frac{3}{2} - \frac{z}{z-1} + \frac{1}{2} \frac{z}{z-2}$$

since ROC: $|z| < 1$, so left sided sequence

so,

$$x[n] = 2\delta[n+1] + \frac{3}{2}\delta[n] + (1)^n u[-n-1] - \frac{1}{2}(2)^n u[-n-1]$$

8L Ba

$$X(z) = \frac{1}{1 - 0.8z^{-1} + 0.12z^{-2}}$$

ROC: $|z| > 0.6$, $|z| < 0.2$, $0.2 < |z| < 0.6$

$$X(z) = \frac{z^2}{z^2 - 0.8z + 0.12}$$

$$X(z) = \frac{z}{(z - 0.6)(z - 0.2)}$$

$$\text{so, } A = \frac{z}{z-0.2} \Big|_{z=0.6} = \frac{3}{2} \quad B = \frac{z}{z-0.6} \Big|_{z=0.2} = -\frac{1}{2}$$

$$X(z) = \frac{3}{2} \frac{1}{z-0.6} - \frac{1}{2} \frac{1}{z-0.2}$$

$$X(z) = \frac{3}{2} \frac{z}{z-0.6} - \frac{1}{2} \frac{z}{z-0.2}$$

$$= \frac{3}{2}$$

a. $|z| > 0.6$: RSS : $\frac{3}{2} (0.6)^n u[n] - \frac{1}{2} (0.2)^n u[n]$

b. $|z| < 0.2$: LSS : $-\frac{3}{2} (0.6)^n u[-n-1] - \frac{1}{2} (0.2)^n u[-n-1]$

c. $0.2 < |z| < 0.6$: Ring shaped: both sided : $-\frac{3}{2} (0.6)^n u[-n-1] + \frac{1}{2} (0.2)^n u[n]$

80 Bh

$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}}$$

$$= \frac{z^2 + 2z + 1}{z^2 - 0.75z + 0.125}$$

$$H(z) = \frac{(z+1)^2}{z(z-0.5)(z-0.25)}$$

$$A = \frac{(z+1)^2}{(z-0.5)(z-0.25)} \Big|_{z=0} \quad B = \frac{(z+1)^2}{z(z-0.25)} \Big|_{z=0.5} \quad C = \frac{(z+1)^2}{z(z-0.5)} \Big|_{z=0.25}$$

$$= 8 \quad = 18 \quad = -25$$

$$H(z) = \frac{8}{z} + \frac{18}{z-0.25} - \frac{25}{z-0.5}$$

$$H(z) = 8 + \frac{18z}{z-0.25} - \frac{25z}{z-0.5}$$

we have: $0.25 < |z| < 0.5$, both sided:

$$h(n) = 8\delta[n] + 18(0.25)^n u[-n-1] - 25(0.5)^n u(n)$$

80 Ba

$$X(z) = 1+z, \text{ same as above}$$

79 Bh, 78 Bh

same as 82 Bh

79 Ba,

same as 78 Bh

only new questions:

76 Ash.

Convolution Property of Z-transform:

The convolution in time domain property of Z-transform states that:
Z-transform of 2 discrete time sequences is equal to multiplication of their Z-transforms.

i.e.,

$$x_1(n) \xrightarrow{ZT} X_1(z); \text{ROF} = R_1$$

$$x_2(n) \xrightarrow{ZT} X_2(z); \text{ROF} = R_2$$

Then,

$$x_1(n) * x_2(n) \xrightarrow{ZT} X_1(z) X_2(z); \text{ROF} = R_1 \cup R_2$$

$$X(z) = \frac{z}{(z-0.6)(z+0.5)^2}$$

$$X(z) = \frac{A}{z-0.6} + \frac{B}{z+0.5} + \frac{Cz}{(z+0.5)^2}$$

$$A = \left. \frac{z}{(z+0.5)^2} \right|_{z=0.6} = \frac{60}{121}$$

$$B = \left. \frac{z}{z-0.6} \right|_{z=-0.5} = \frac{5}{11}$$

$$C = \left. \frac{d}{dz} \frac{z}{z-0.6} \right|_{z=-0.5} = \frac{(z-0.6) - z}{(z-0.6)^2} \bigg|_{z=-0.5} = \frac{-60}{121}$$

$$X(z) = \frac{60}{121} \times \frac{1}{z-0.6} + \frac{5}{11} \times \frac{1}{(z+0.5)^2} - \frac{60}{121} \times \frac{1}{z+0.5}$$

$$X(z) = \frac{60}{121} \times \frac{z}{z-0.6} + \frac{5}{11} \times \frac{z+0.5}{(z+0.5)^2} - \frac{60}{121} \times \frac{z}{z+0.5}$$

$|z| > 0.6$ (RSS)

$$x[n] = \frac{60}{121} \times (0.6)^n u[n] + \frac{5}{11} n (0.5)^{n-1} u[n-1] - \frac{60}{121} (0.5)^n u[n]$$

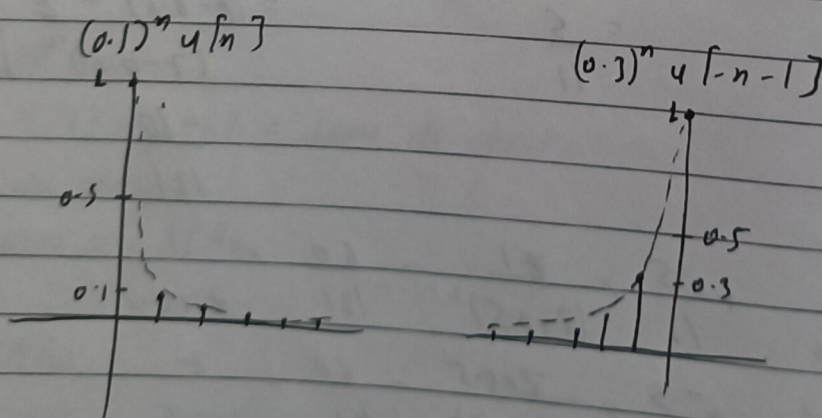
75 (h)

6) Properties of Z-transform's ROC

- The ROC is a ring or a disk in the Z plane centred at origin
 $0 \leq r_1 < |z| < r_2 < \infty$ R $\rightarrow$ Right sided, L $\rightarrow$ Left sided
- Fourier transform of $x[n]$ converges absolutely if and only if the ROC of the ZT of $x[n]$ includes the unit circle
- ROC doesn't contain any poles
- If $x[n]$ is finite duration sequence, then ROC is entire Z-plane, except possibly at $z=0$ or $z=\infty$
- If $x[n]$ = RSS that is zero after some N , ROC extends outward from the outermost pole in $X(z)$ to $z=\infty$
- If $x[n]$ = LSS, that is zero before some N , ROC extends inward from the innermost pole in Z-plane $X(z)$ to $z=0$
- Two sided sequence will have ROC as annulus in Z plane, bounded on interior/exterior by a pole & not containing any pole
- ROC is a simply connected region.

R.O.C of :

$$x[n] = (0.1)^n u[n] + (0.3)^n u[-n-1]$$



let $x_1[n] = [0.1]^n u[n]$

$x_2[n] = [0.3]^n u[-n-1]$

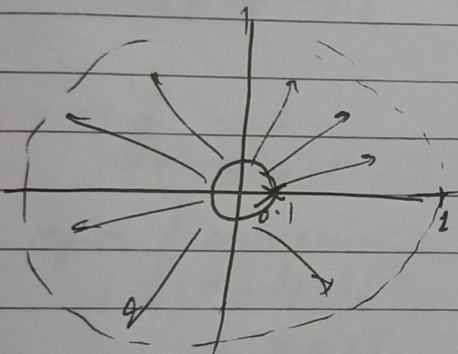
& $x[n] = x_1[n] + x_2[n]$

$X_1(z) = \sum_{n=-\infty}^{\infty} x_1[n] z^{-n}$

$= \frac{1}{1 - 0.1z^{-1}} = \frac{z}{z - 0.1}$, for convergence:

$|0.1z^{-1}| < 1$
 $|10z| > 1$
 $|z| > \frac{1}{10}$

* ROC for $x_1[n]$:



$X_2(z) = \sum_{n=-\infty}^{\infty} x_2[n] z^{-n}$

$= \sum_{n=-\infty}^{-1} (0.3)^n u[-n-1] z^{-n}$

$= \sum_{n=1}^{\infty} (0.3)^{-n} z^{nn}$

$= \sum_{n=1}^{\infty} (0.3^{-1} z)^n$

$= \left(\sum_{n=0}^{\infty} (0.3^{-1} z)^n \right) - 1$

$= \frac{1}{1 - 0.3^{-1} z} - 1 = \frac{z}{z - 0.3^{-1}} = \frac{z}{z - 0.3^{-1}}$

$$= \frac{1}{1-0.3z^{-1}} - 1$$

$$= \frac{1 - 1 + 0.3z^{-1}z}{1-0.3z^{-1}z}$$

$$= \frac{0.3z^{-1}z}{1-0.3z^{-1}z}$$

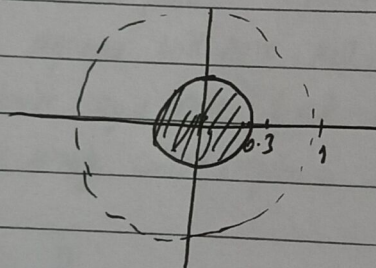
zero at $z=0$.

for ROC:

$$|0.3z^{-1}z| < 1$$

$$|z| < 0.3$$

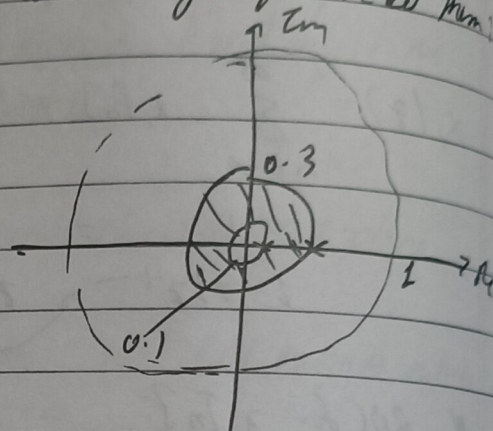
no ROC:



zero at $z=0, 0.2$

poles at $z=1, 0.3$

no common region of ROC (but "them")



also,

$$X(z) = X_1(z) + X_2(z)$$

$$= \frac{z}{z-0.1} - \frac{0.3z^{-1}}{1-0.3z^{-1}z}$$

$$= \frac{z}{z-0.1} - \frac{z}{0.3-z}$$

$$= \frac{z}{z-0.1} + \frac{z}{z-0.3}$$

$$= \frac{z^2 - 0.3z + z^2 - 0.1z}{(z-1)(z-0.3)}$$

$$= \frac{2z^2 - 0.4z}{(z-1)(z-0.3)}$$

$$= \frac{2z(z-0.2)}{(z-1)(z-0.3)}$$

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$$RVC: x[n] = (0.6)^n u[n] + (0.25)^n u[n]$$

so,

$$x_1 = (0.6)^n u[n]$$

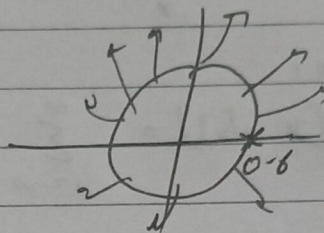
$$X_1 = \frac{z}{z - 0.6}$$

and: for convergence:

$$|0.6 z^{-1}| < 1$$

$$|0.6| < z$$

$$|z| > 0.6$$

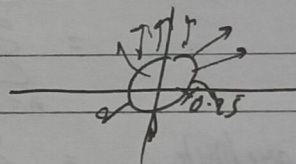
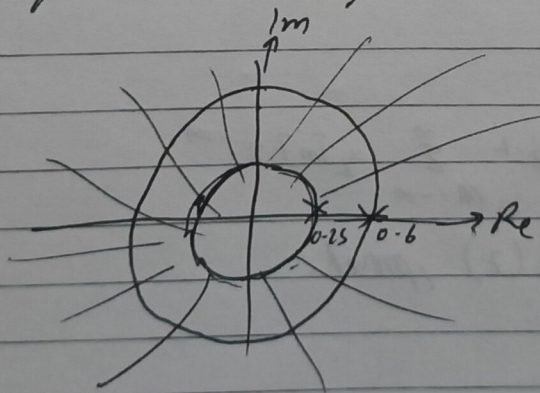


$$X_2 = \frac{z}{z - 0.25}$$

for region of convergence,

$$|0.25 z^{-1}| < 1$$

$$\therefore z > 0.25$$

 $\therefore$ RVC of the combined system:

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Derive convolution property

$$x_1[n] * x_2[n] = \sum_{k=-\infty}^{\infty} x_1[k] x_2[n-k]$$

Z transform:

$$Z[x[n]] = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$\begin{aligned} \therefore Z[x_1[n] * x_2[n]] &= \sum_{n=-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} x_1[k] x_2[n-k] \right) z^{-n} \\ &= \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} x_1[k] x_2[n-k] z^{-k} z^{-(n-k)} \end{aligned}$$

if sums are finite,
they can be interchanged

$$= \sum_{k=-\infty}^{\infty} x_1[k] z^{-k} \sum_{n=-\infty}^{\infty} x_2[n-k] z^{-(n-k)}$$

let $n-k = m$

at $n = \infty$, $n = -\infty$

$m = \infty$ $m = -\infty$

$$= \sum_{k=-\infty}^{\infty} x_1[k] z^{-k} \sum_{m=-\infty}^{\infty} x_2[m] z^{-m}$$

$$= X_1(z) X_2(z), \text{ proved.}$$